

LAPLACE OPERATORS OF INFINITE-DIMENSIONAL LIE ALGEBRAS AND THETA FUNCTIONS

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This is a presentation on *V.G. Kac's* paper [Ka84] of the same name.

The Problem: Until this paper, the generalized Casimir operator was the only known element of the center of a completion of the enveloping algebra of a Kac-Moody algebra $\mathfrak{g}$.

The Conjecture: Let K be the complexified tits cone and W the Weyl group of $\mathfrak{g}$. It was proved by Looijenga in 1980 that some θ -functions on K separate the W -orbits in K . Then it was conjectured in [DGK] (in their introduction, p. 93) that the image of some Harish-Chandra homomorphism should contain these θ -functions.

The Main Result: The paper constructs some completion of $U(\mathfrak{g})$ with a center large enough to separate the W -orbits.

I will be presenting sections 1 through 4 that succeed in setting up the completion space, defining the Laplace operators and proving the theorem that when they commute with the action of $\mathfrak{g}$ and act on a highest weight module $M(\Lambda)$, they commute with everything and act purely as a scalar multiplier.

LIST OF NOTATIONS

A	symmetrisable $n \times n$ cartan matrix of rank r with entries $(a_{i,j})_{i,j=1}^n$
$\mathfrak{h}_{\mathbb{R}}$	a vector space over $\mathbb{R}$ of dimension $2n - r$
Π	the set of simple roots $(\alpha_i)_{i=1}^n$
$\Pi^{\vee}$	the set of simple co-roots $(\alpha_i^{\vee})_{i=1}^n$
Q	the group generated by the simple roots in $\mathfrak{h}_{\mathbb{R}}^*$
Δ_+	The set of positive roots
$\tilde{\Delta}_+$	The multiset of positive roots with multiplicity being dimension of weight space.
$\mathfrak{g}$	Kac-Moody lie algebra associated with A .
$\mathfrak{h}$	$\mathfrak{h}_{\mathbb{R}} \otimes \mathbb{C}$

PART 1: PRELIMINARIES

The setup is slightly different from the setup we used in class. The realisation of A is taken over $\mathbb{R}$ and denoted by $(\mathfrak{h}_{\mathbb{R}}, \Pi, \Pi^{\vee})$.

Given a $\Lambda \in \mathfrak{h}^*$, we define **universal Verma module** as $M(\lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_{\lambda}$. I find it easier to think of it as $M(\Lambda)$ as $U(n^-)v_{\Lambda}$ i.e as $U(n^-)$ modules they are isomorphic, so we can think of it as generated by v_{Λ} as a $U(n^-)$ module. We use $J(\Lambda)$ to denote the unique maximal submodule of $M(\Lambda)$. It exists because $M(\Lambda) = M(\Lambda)_{\Lambda} \oplus \bigoplus_{\beta > 0} M(\Lambda)_{\Lambda - \beta}$ and every proper submodule is contained in the second summand. After this we can uniquely define the irreducible quotient which is $L(\Lambda) := M(\Lambda)/J(\Lambda)$.

PART 2: THE CONTRAVARIANT FORM:

For a given element $\beta \in Q_+$, the set of partition functions is denoted by $\text{Par } \beta$ and is defined as the set of maps $k : \tilde{\Delta}_+ \rightarrow \mathbb{Z}_+$ (expressed as $\alpha \mapsto k_\alpha$) such that $\beta = \sum_{\alpha \in \Delta_+} k_\alpha \alpha$. The space of all such partition functions across the entire positive root lattice is defined by taking the union over all $\beta \in Q_+$. This is denoted as:

$$\text{Par} = \cup_{\beta \in Q_+} \text{Par } \beta$$

For each positive root $\alpha \in \Delta_+$, we choose a specific basis $\{f_\alpha^{(i)}\}_i$ for the corresponding negative root space $\mathfrak{g}_{-\alpha}$. Once these individual bases are chosen, we take the union of all of them and arrange them into an ordered sequence denoted as $F_1, F_2, \dots$.

For a partition $k \in \text{Par } \beta$ we define an ordered product of these basis elements. We denote by $F^k = F_1^{k_1} F_2^{k_2} \dots \in U(\mathfrak{n}_-)$. According to the Poincaré-Birkhoff-Witt (PBW) theorem, the collection of all such terms F^k for a fixed partition sequence $k \in \text{Par } \beta$ forms a basis for the vector space $U(\mathfrak{n}_-)_{-\beta}$.

Now we construct the form. We fix $\lambda \in \mathfrak{h}^*$.

We first build it **locally** i.e. we define it first on $U(\mathfrak{n}_-)_{-\beta}$. let $k, m \in \text{Par}(\beta)$ and define $B_\beta^\lambda(F^k, F^m) = c$ where c is the coefficient that appears in the equation:

$$\sigma(F^k)F^m(v_\lambda) = c \cdot (v_\lambda)$$

Now we handle the **global** case. The global bilinear form B on the module $M(\lambda)$ is defined by decreeing that:

- (1) local forms govern the pairings within a single weight space

$$B(F^k v_\lambda, F^m v_\lambda) = B_\beta^\lambda(F^k, F^m)$$

- (2) distinct weight spaces are mutually orthogonal: for $\mu \neq \nu$

$$B(M(\lambda)_\mu, M(\lambda)_\nu) = 0$$

Lemma 1. *The kernel of this bilinear form is precisely $J(\Lambda)$*

Proof. The contravariance property of B yields : $B(g \cdot x, y) = B(x, \sigma(g) \cdot y)$. This makes the kernel of B a $\mathfrak{g}$ submodule. It is a proper submodule as $B(v_\Lambda, v_\Lambda) = 1 \neq 0$. Let $N \subsetneq M(\Lambda)$ be any proper submodule. Because $M(\Lambda)$ is a highest weight module, N decomposes into weight spaces and does not contain the highest weight vector v_Λ . Consequently, the Λ -weight space of N is empty, meaning no element of N has a component of weight Λ . Let $n \in N$. We want to show $B(n, y) = 0$ for all $y \in M(\Lambda)$. Since v_Λ generates $M(\Lambda)$, any $y \in M(\Lambda)$ can be written as $y = u \cdot v_\Lambda$ for some $u \in U(\mathfrak{n}_-)$. Using the contravariance of B :

$$B(n, y) = B(n, u \cdot v_\Lambda) = B(\sigma(u) \cdot n, v_\Lambda)$$

. Because N is a submodule, $\sigma(u) \cdot n \in N$. As established, N has no weight components in the Λ -weight space, and since distinct weight spaces are orthogonal under B , pairing any element of N with v_Λ yields zero. Thus, $B(\sigma(u) \cdot n, v_\Lambda) = 0$. $\square$

A Verma module $M(\Lambda)$ is irreducible if and only if $J(\Lambda) = 0$. Consequently, $M(\Lambda)$ is irreducible if and only if the contravariant form is non-degenerate, which occurs exactly when $\det B_\beta^\lambda \neq 0$ for all $\beta > 0$.

The next lemma is 'lemma 1' in Kac's paper. It goes back to the paper [KK79].

Lemma 2. Choose $\rho \in \mathfrak{h}^*$ such that $\langle \rho, \alpha_i^\vee \rangle = 1$ for $i = 1, \dots, n$. For any positive root $\alpha \in Q_+$, we define the affine linear function on the dual Cartan space $\mathfrak{h}^*$:

$$T_\alpha(\lambda) = 2(\lambda + \rho|\alpha) - (\alpha|\alpha)$$

(i) The determinant of the contravariant bilinear form matrix evaluates to:

$$\det B_\beta^\lambda = c \prod_{\alpha \in \Delta_+} \prod_{n=1}^{\infty} T_{n\alpha}(\lambda)^{\text{card Par}(\beta - n\alpha)}$$

where c is a nonzero constant entirely independent of λ .

(ii) Let $\beta = n\alpha$, where n is a positive integer and $\alpha \in \Delta_+$ and let $\Lambda \in \mathfrak{h}^*$ be such that $T_\beta(\Lambda) = 0$. Then the g -module $M(\Lambda)$ contains a submodule isomorphic to $M(\Lambda - \beta)$.

(iii) Let $\beta = n\alpha$ and Λ be as in (ii), and assume, furthermore, that α is a real root ($\alpha \in \Delta_+^{re}$) and that $T_{n\gamma}(\Lambda) \neq 0$ for all positive roots $\gamma \in \Delta_+ \setminus \{\alpha\}$. Then $J(\Lambda) \cong M(\Lambda - n\alpha)$.

when is $\det B_\beta^\lambda = 0$?

It is 0 on the infinite sets of hyperplanes defined by $T_{n\alpha}(\lambda) = 0$. These hyperplanes are given by the equation:

$$2(\Lambda|\alpha) = n(\alpha|\alpha) - 2(\rho|\alpha)$$

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PART 3: CONSTRUCTING THE COMPLETION

One has

$$\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$$

and therefore

$$U(\mathfrak{g}) = U(\mathfrak{n}_-) \otimes U(\mathfrak{h}) \otimes U(\mathfrak{n}_+)$$

By the Poincaré-Birkhoff-Witt (PBW) theorem, the standard elements of $U(\mathfrak{g})$ can be written in the form $F^k \phi_{k,m} \sigma(F^m)$, where $\phi_{k,m}$ belongs to $S(\mathfrak{h})$.

Remark 1: Instead of polynomials, Kac wants to allow rational functions that have poles at specific locations. He defines a set L , which is the union of affine hyperplanes in the of the form $T_{n\alpha}(\lambda + \gamma) = 0$. In this equation, n is a positive integer, $\alpha \in \Delta_+$ (positive roots), and $\gamma \in Q$. He then defines $\mathfrak{F}$ as the algebra of complex-valued functions that are well-defined outside of these hyperplanes – meaning functions defined on the domain $\mathfrak{h}^ \setminus L$.*

L is defined as the union of all affine hyperplanes in $\mathfrak{h}^*$ of the form $T_{n\alpha}(\lambda + \gamma) = 0$.

$\mathfrak{F}$ is the algebra of complex-valued functions that are well-defined on the domain $\mathfrak{h}^* \setminus L$

We take the **free product** of $U(\mathfrak{g})$ and $\mathfrak{F}$ that is $U(\mathfrak{g}) * \mathfrak{F}$ and then quotient by the two sided ideal generated by -

(1)

$$h - j(h)$$

for all $h \in \mathfrak{h}$

This forces the old Cartan polynomials (h) inside $U(\mathfrak{g})$ to be strictly identified with their corresponding function versions ($j(h)$) inside $\mathfrak{F}$

(2)

$$e_\alpha \varphi(\lambda) - \varphi(\lambda) e_\alpha - (\varphi(\lambda + \alpha) - \varphi(\lambda)) e_\alpha = 0$$

where $\varphi(\lambda) \in \mathfrak{F}$ and $e_\alpha \in \mathfrak{g}_\alpha$.

Setting this to zero we get:

$$e_\alpha \varphi(\lambda) = \varphi(\lambda + \alpha) e_\alpha$$

The set of ordered monomials $\mathcal{B} = \{F^k \phi_j E^m\}$ forms a basis for $U_{\mathfrak{F}}(\mathfrak{g})$ where ϕ_j runs through a basis of $\mathfrak{F}$.

We introduce the notation $\beta(k)$ to denote the weight of a partition k :

$$\beta(k) = \sum_{\alpha \in \tilde{\Delta}_+} k(\alpha) \alpha$$

We define the height function, $\text{ht} : Q \rightarrow \mathbb{Z}$, defined as the linear form:

$$\text{ht} \left(\sum_{i=1}^n c_i \alpha_i \right) = \sum_{i=1}^n c_i$$

Using this we get:

$$\deg(F^k) = - \sum_{\alpha \in \tilde{\Delta}_+} k(\alpha) \text{ht}(\alpha) = -\text{ht}(\beta(k))$$

It follows that the degree of a general basis element $F^k \varphi_{k,m} \sigma(F^m)$ evaluates precisely to the difference in heights:

$$\text{ht}(\beta(m)) - \text{ht}(\beta(k)) = \text{ht}(\beta(m) - \beta(k))$$

In the principal gradation of $U(\mathfrak{g})$ (and respectively $U_{\mathfrak{F}}(\mathfrak{g})$), the part of degree j is the $\mathbb{C}$ -vector space generated by finite sums of elements of the form $F^k \varphi_{k,m} \sigma(F^m)$, where $\varphi_{k,m} \in \mathfrak{F}$ and:

$$\text{ht}(\beta(m) - \beta(k)) = j$$

Kac then defines the completion, $\tilde{U}_{\mathfrak{F}}(\mathfrak{g})_j$, as the space comprising all infinite sums of such elements that satisfy this height condition, specifically:

$$\tilde{U}_{\mathfrak{F}}(\mathfrak{g})_j = \prod_{\substack{k,m \\ \text{ht}(\beta(m) - \beta(k)) = j}} F^k \varphi \sigma(F^m)$$

Now that the completion is done we restrict to those modules V in category $\mathcal{O}$ for which $P(V) \subset \mathfrak{h}^* - L$, where $P(V)$ denotes the set of weights occurring in the module as it is in category $\mathcal{O}$. We can extend the action of the standard enveloping algebra $U(\mathfrak{g})$ to the completed algebra $\tilde{U}_{\mathfrak{F}}(\mathfrak{g})$. For the elements $\varphi \in \mathfrak{F}$, Kac defines the action as

$$\varphi(v_\lambda) = \varphi(\lambda) v_\lambda$$

In particular Verma modules with highest weight outside L satisfy all the conditions and section 3 ends by showing the action of the completion on these modules.

PART 4: THE COMPLETED ENVELOPING ALGEBRA

The goal of section 4 is to establish the structure of the center of our constructed completion algebra. The section begins by denoting the center of the algebra $\hat{U}_{\mathfrak{F}}(\mathfrak{g})$ as $Z_{\mathfrak{F}}$.

We finally get to first theorem of the paper, formally it is stated as:

Theorem 1 : Given a function $\varphi \in \mathfrak{F}$, there exists a unique element z in the center $Z_{\mathfrak{F}}$ of the algebra $\hat{U}_{\mathfrak{F}}(\mathfrak{g})$ such that :

$$z_\varphi = \sum_{\beta \in Q^+} \sum_{k,m \in \text{Par} \beta} F^k \varphi_{k,m} \sigma(F^m)$$

subject to the condition that:

$$\varphi_{0,0} = \varphi$$

Proof. We will use the result that if z acts as a scalar on each module $M(\Lambda)$ with $\Lambda \in \mathfrak{h}^* \setminus L$, then $z \in Z_{\mathfrak{g}}$. A proof of this can be found in the appendix.

Using the following three things:

- (1) z acts as a scalar
- (2) Induction on the heights of elements in Q_+
- (3) We have a method to write z using the basis we constructed (*It is the double sum given above*)

We will try to explicitly calculate $\varphi_{k,m}$ that appear in the basis.

Fix $\beta \in Q^+ - \{0\}$ and then for each $\{\gamma \in Q^+ \mid \gamma \leq \beta\}$ we let G_γ represent the matrix of the operator $\sum_{k,m \in \text{Par}\gamma} F^k \varphi_{k,m} \sigma(F^m)$ acting on the space $M(\lambda)_{\lambda-\gamma}$ in the basis $F^s(v_\lambda)$.

Remark : By the Poincaré-Birkhoff-Witt theorem, the F^s terms form a basis for the vector space $U(\mathfrak{n}_-)_\beta$, the vectors $F^s(v_\lambda)$ are linearly independent and span the entire weight space $M(\lambda)_{\lambda-\beta}$ hence they form a natural basis which we are fixing in order to have a matrix representation for the linear map.

Since we are inducting on elements of Q^+ so by our inductive assumption we already know all the matrices of the form G_γ for any $\gamma < \beta$. let $\Phi_\beta = (\varphi_{k,m})$ for $k, m \in \text{Par}\beta$, as $\text{Par}\beta$ is a finite set we get a $n \times n$ matrix for some n .

The premise is that the central element z must act exactly as the scalar $\varphi(\lambda)$ on the specific weight space $M(\lambda)_{\lambda-\beta}$. This means the total operator's matrix must perfectly equal $\varphi(\lambda)I$.

z is an infinite sum over all roots $\gamma \in Q^+$. When we apply this operator to the specific weight space $M(\lambda)_{\lambda-\beta}$, the infinite sum naturally breaks into three distinct parts:

- (1) lower weights ($\gamma < \beta$) : The action on our weight space is represented by the sum $\sum_{\gamma < \beta} G_\gamma$
- (2) higher weights ($\gamma > \beta$) : annihilates our weight space
- (3) current weight ($\gamma = \beta$)

To see how we get $G_\beta = \Phi_\beta B_\beta$ we look at how it acts on v_λ :

$$(1) \quad \left(\sum_{k,m \in \text{Par}\beta} F^k \varphi_{k,m} \sigma(F^m) \right) F^s(v_\lambda)$$

$$(2) \quad \sum_{k,m} F^k \varphi_{k,m} (\sigma(F^m) F^s(v_\lambda))$$

$$(3) \quad \sigma(F^m) F^s(v_\lambda) = B_\beta(F^m, F^s) v_\lambda$$

We denote this scalar by $b_{m,s}$ now onwards.

$$(4) \quad \sum_{k,m} F^k \varphi_{k,m} (b_{m,s} v_\lambda)$$

$$(5) \quad \sum_k \left(\sum_m \varphi_{k,m}(\lambda) b_{m,s} \right) F^k(v_\lambda)$$

This is a linear combination of basis vectors with the coefficient for the k -th basis vector being exactly the inner sum: $\sum_m \varphi_{k,m} B_\beta(F^m, F^s)$. B_β is the matrix with entries $B_\beta(m, s) = B_\beta(F^m, F^s)$. If you multiply the matrix Φ_β by the matrix B_β , the entry in (k, s) is defined precisely as:

$$\sum_m \Phi_\beta(k, m) B_\beta(m, s)$$

This yields the beautiful and clean factorization:

$$G_\beta = \Phi_\beta B_\beta$$

Using the above factorisation we get the equation:

$$\Phi_\beta B_\beta + \sum_{\gamma < \beta} G_\gamma = \varphi(\lambda) I$$

Because the weight λ is chosen to be outside the hyperplanes L , the determinant of the contravariant form B_β is non-zero ($\det B_\beta \neq 0$). Consequently by inverting B_β we uniquely determine the coefficient matrix Φ_β at λ .

Following this proof the paper remarks $H : Z_{\mathfrak{F}} \rightarrow \mathfrak{F}$ is a homomorphism and that theorem 1 proves it is an isomorphism. We first show that this map is a morphism of algebras, the proof for which is from [Hu72], §23.3. For this we define the evaluation map $\varepsilon_\lambda : \mathfrak{F} \rightarrow \mathbb{C}$, which evaluates a given function at λ then we have

$$\forall \lambda \in \mathfrak{h}^*, \forall z \in Z_{\mathfrak{F}} \quad \varepsilon_\lambda(H(z)) = \chi_\lambda(z)$$

where $\chi_\lambda(z)$ is the character by which by which the center acts on the Verma modules.

$$\varepsilon_\lambda(H(z z')) = \chi_\lambda(z z') = \chi_\lambda(z) \chi_\lambda(z') = \varepsilon_\lambda(H(z)) \varepsilon_\lambda(H(z')) = \varepsilon_\lambda(H(z) H(z'))$$

Because this equality holds true for all $\lambda \in \mathfrak{h}^*$, it proves that the functions evaluate identically everywhere, yielding the final conclusion:

$$H(z z') = H(z) H(z')$$

The fact it is surjective is tautological from the statement of theorem 1 and injectivity follows from uniqueness. $\square$

APPENDIX A. DEFERRED PROOFS

This section contains the proof omitted from the main text.

Proposition 1: If $z \in \mathcal{U}$ acts as a scalar on Verma modules $M(\lambda)$ for $\lambda \in \mathfrak{h}^* - L$ then $x = [u, z]$ must act as the zero operator on $M(\lambda)$.

Proof. Let $\Lambda \in \mathfrak{h}^* - L$ and let $v \in M(\Lambda)$ be an arbitrary vector. By the hypothesis, z acts as a scalar c_Λ on the entire module $M(\Lambda)$. Therefore:

$$z \cdot m = c_\Lambda m \quad \forall m \in M(\Lambda)$$

The same scalar working for the entire module comes from the fact that ring of $\mathfrak{g}$ -module homomorphisms of $M(\Lambda)$ is just $\mathbb{C}$. We evaluate the action of the commutator $u = [z, x]$ on the vector v :

$$[z, x] \cdot v = (zx - xz) \cdot v = z \cdot (x \cdot v) - x \cdot (z \cdot v)$$

- (1) $z \cdot (x \cdot v) = c_\Lambda (x \cdot v)$
- (2) $x \cdot (z \cdot v) = x \cdot (c_\Lambda v) = c_\Lambda (x \cdot v)$

Substituting these into the commutator expression one obtains:

$$[z, x] \cdot v = c_\Lambda(x \cdot v) - c_\Lambda(x \cdot v) = 0$$

Since $[z, x] \cdot v = 0$ for all $v \in M(\Lambda)$, the element $[z, x]$ acts as the zero operator on $M(\Lambda)$. $\square$

Proposition 2: If $z \in \mathcal{U}$ acts as a scalar on each module $M(\Lambda)$ (with $\Lambda \in \mathfrak{h}^* \setminus L$), then for all $x = [u, z]$ (with u in the completion) we show $x = 0$.

Proof. We start with the assumption that $z \in \mathcal{U}$ acts as a scalar on verma modules $M(\lambda)$. Consequently, any commutator $x = [u, z]$ must act as the zero operator on $M(\lambda)$.

$$I := \{x \in \mathcal{U} \mid x|_{M(\Lambda)} = 0 \ \forall \ \Lambda \in \mathfrak{h}^* - L\}$$

It is enough to show $I = 0$.

We first define a grading on $\mathcal{U}$. we fix $h \in \mathfrak{h}$ such that $\alpha_i(h) = 1$. Such an h exists because for any vector space V over a field K and n - linearly independent elements in $\beta_i \in V^*$ for $i = 1$ to n the map $(\beta_1, \beta_2, \dots, \beta_n) : V \rightarrow K^n$ is a surjective map.

The action of $ad(h)$ on chevalley generators is diagonal by the eigenvalues 0 on $\mathfrak{h}$, 1 on e_i and -1 on f_i . This action extends to $\mathcal{U}$ and we get the grading:

$$\mathcal{U} = \bigoplus_{j \in \mathbb{Z}} \mathcal{U}_j$$

Since the ideal I is stable under this adjoint action, it splits into eigenspaces: $I = \bigoplus (I \cap \mathcal{U}_j)$. Because of this, it is sufficient to pick an arbitrary element $x \in I$ and assume it belongs to a specific homogeneous degree j . If we prove $x = 0$ for an arbitrary homogeneous component, we prove $I = 0$. Using the PBW theorem and the basis we fixed above we get:

$$x = \sum \sigma(E^k) \varphi_{k,m} E^m : \text{where } \text{ht}(\beta(m) - \beta(k)) = j$$

We prove that the $m = 0$ terms vanish as follows:

$$xv_\lambda = \sum_{|k|=-j} F^k \varphi_{k,0}(\lambda) v_\lambda = 0$$

Because the Verma module $M(\lambda)$ is free as a $U(\mathfrak{n}_-)$ module, the vectors $F^k v_\lambda$ are linearly independent. For the sum to equal zero, every scalar coefficient $\varphi_{k,0}(\lambda)$ must be zero. Since this holds for all λ , the polynomials $\varphi_{k,0}$ are identically 0.

Now assume, for the sake of contradiction, that $x \neq 0$. Since the terms where $m = 0$ vanish, there must exist some $m \neq 0$ such that $\varphi_{k,m} \neq 0$. Choose an index m such that its height $r = \text{ht}(\beta(m))$ is strictly minimal among all non-zero terms. Let $\beta = \beta(m)$ be the weight of this element.

The elements $\sigma(E^{m'})$ with weight $\beta(m') = \beta$ form a basis for the weight space $\mathcal{U}(\mathfrak{n}_-)_{-\beta}$. The contravariant form B_β^λ on this weight space is non-degenerate. Let $\{Y^{m'}\}$ be the dual basis of $\mathcal{U}(\mathfrak{n}_-)_{-\beta}$ relative to B_β^λ , uniquely defined by the property:

$$B_\beta^\lambda(\sigma(E^p), Y^m) = \delta_{p,m}$$

Apply x to the vector $Y^m v_\lambda \in M(\lambda)$. Because $x \in I$, it annihilates the entire module, yielding:

$$xY^m v_\lambda = \sum_k \sum_p F^k \varphi_{k,p} E^p Y^m v_\lambda = 0$$

We evaluate the action of E^p on $Y^m v_\lambda$ for each term in the sum, relying on the minimality of $r = |m|$:

- (1) If $|p| > r$, the weight of $E^p Y^m v_\lambda$ is strictly greater than λ . Such a weight space does not exist in $M(\lambda)$, so the term is 0
- (2) If $|p| = r$ but $\beta(p) \neq \beta(m)$, the weights do not match, and by weight space decomposition, the term is 0.
- (3) If $\beta(p) = \beta(m)$, the action reduces to the evaluation of the contravariant form. By our choice of the dual basis, $E^p Y^m v_\lambda = B_\beta^\lambda(\sigma(E^p), Y^m) v_\lambda = \delta_{p,m} v_\lambda$

Because of the dual basis relations, every term in the expansion vanishes except when $p = m$. The sum collapses entirely to:

$$0 = \sum_{|k|=|m|-j} F^k \varphi_{k,m}(\lambda) v_\lambda$$

By linear independence the coefficients must vanish: $\varphi_{k,m}(\lambda) = 0$ for all λ . This contradicts our choice, where m was specifically chosen as an index for which $\varphi_{k,m} \neq 0$. Therefore, our assumption that $x \neq 0$ must be false. We conclude $x = 0$. $\square$

The above proof was given to me by **Prof Patrick Polo** and is similar to the proof in [CI84] (*Th. 2.6, assertion (i)*). Prof Patrick gave a course on Kac-Moody Lie Algebras in **CMI** and this writeup was written as part of the evaluation for the course and overseen by him. Needless to say, any errors that you find are mine alone.

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